

Cloudflare's 11.5 Tbps DDoS mitigation

I selected this because it's one of the largest publicly-reported attacks in 2025, had broad implications for Internet infrastructure defenders, and was well-covered by reputable sources.

Target

Cloudflare disclosed mitigation of a multi-terabit DDoS aimed at its network / customers (attack surfaced publicly 2 Sep 2025). The attack was blocked at Cloudflare's edge before causing widespread outages to their protected customers.

Technology used by attackers

- **Volumetric flooding (Layer 3/4):** Extremely high aggregate bandwidth (reported peak ~11.5 Tbps) and very large packet rates.
- **Multi-vector behavior:** Attackers combine different flood types (UDP, SYN/ACK, possibly reflection/amplification) to maximize bandwidth and packet-pressure.
- **Botnet orchestration:** Sizable distributed botnet of compromised devices (IoT, servers) able to generate many simultaneous flows and high throughput.
- **Rapid bursts:** Short, intense bursts designed to overwhelm capacity and probe mitigation thresholds.
Cloudflare and other mitigators noted both bandwidth and packet-rate stressors which require different mitigation tactics (rate-limiting vs. capacity scrubbing).

Attacker motive

Public reporting did not attribute the attack to a named actor. Typical motives for attacks of this type include:

- **Capability demonstration** (showing off an attacker/botnet's power).
- **Test/probe** of mitigation capacity of major providers.
- **Disruption or coercion** (service disruption, extortion) — though no extortion demand was publicly linked to this specific event.
Because major mitigation vendors were the target, testing or bragging is a plausible motive.

Overall impact

- **Direct service impact:** Cloudflare's network absorbed and mitigated the traffic; many customers likely experienced no disruption thanks to edge scrubbing. Where customers relied on smaller providers with less mitigation capacity, there's higher risk of degradation.
- **Industry impact:** The event underscored that very large volumetric attacks are still increasing in size and that defenders must scale capacity and packet-rate handling. It elevated urgency around global scrubbing capacity, ISP collaboration, and anti-botnet measures.

Defensive strategies that could have mitigated or reduced impact

(These are best practices and technical countermeasures defenders should use.)

1. **Massive distributed scrubbing capacity at the edge**
 - Use globally distributed CDN/edge providers (Cloudflare, Akamai, etc.) to absorb volumetric traffic near source and avoid saturating customer links.
2. **High packet-rate handling & packet-level filtering**
 - Ensure mitigation can handle not only raw bandwidth but also extreme packet per second (PPS) loads (specialized hardware and fast path software).
3. **Upstream ISP coordination and blackholing / selective routing**
 - Collaborate with transit providers for traffic diversion to scrubbing centers rather than dropping legitimate traffic. BGP-based sinkholing / flow redirection to scrubbing facilities helps preserve service availability.
4. **Rate limiting, connection throttling and stateless filters**
 - Apply rate and protocol filters for suspicious flows; use SYN cookies and stateful defense for TCP floods.
5. **DDoS detection & traffic anomaly baselining**
 - Fast detection using behavioral baselines (traffic patterns, geo/AS deviations) to automatically trigger mitigation playbooks.
6. **Botnet disruption and takedowns**
 - Coordinate with law enforcement and CERTs to takedown C2 infrastructure and botnet management servers where feasible.
7. **Zero-trust / application hardening**
 - For application layer risk, require proof-of-work, CAPTCHA, or API tokens to make HTTP floods costly for attackers.
8. **Redundancy & regional failover**
 - Multi-region deployments with failover can reduce impact on end users when a particular path is saturated.

Short recommendations for operators

- Test failover and scrubbing workflows; run tabletop exercises for DDoS response.
- Ensure SLAs with upstream providers include DDoS handling/flow diversion.
- Monitor global threat intel (new botnets, reflection vectors) and tune edge filtering accordingly.